

IDEAS

① MAPS

Dot map

Each dot is a tornado.
Size could be # of deaths, path length, \$ damage etc.
Colour could be EF rating.
Map would have graticules + state boundaries.

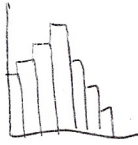
Choropleth map
Each state is coloured using a sequential colour scale, e.g. yellow-green-blue could depict # of deaths, \$ damage, # of tornadoes, etc.

Heat map

To show poverty levels, transmission line voltages, etc.
To be underlaid with a dot map?

② charts (non-map)

Histogram



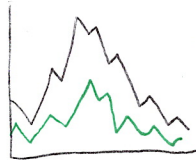
Distribution of # deaths, \$ damage, path length, tornado width

Pie chart



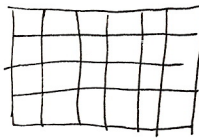
% of tornadoes of different EF ratings

Line chart



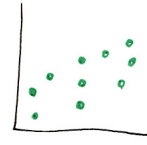
tornadoes per month, year, quarter?
Maybe each line could be a diff. year

Calendar heatmap



Each day is coloured based on how many tornadoes there were

Scatterplot



correlation between \$ damage and EF? or with deaths

Area chart



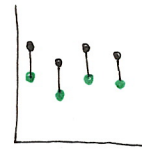
could be an area chart instead?

Pictogram / isotype



Each symbol = a tornado.
A portion of coloured tornadoes = fatal ones?

Dot plot



change in # of tornadoes between 2 different years (or # of deaths, injuries, etc.)

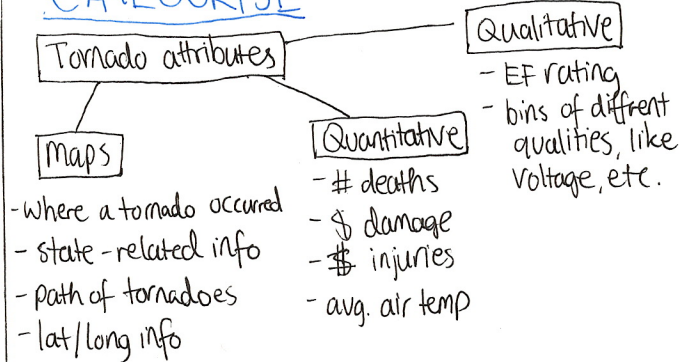
FILTER

② calendar heatmap:
probably not feasible for a year, unless the user can change the month

② pie chart:
A bit boring, could be combined w/ another chart to give it more dimension

② Pictogram:
Probably too many tornadoes to have each represented by one symbol.
Maybe one symbol = 10 tornadoes?

CATEGORISE



Author: Amaris Zhang

Date: 15/9/2025

Sheet: 1

Task: Planning charts and choosing ones for the visualisation

QUESTION

- Are there not enough qualitative attributes, and too many quantitative?
- will the map be too crowded with data points or info?
- should I restrict some of the data, e.g. only tornadoes w/ deaths?

COMBINE + REFINE

- calendar heatmap could be made into a line chart to better visualise changes over time
- pie chart could be instead an area chart; this also shows change over time
- pictogram could be instead a map; this takes up less space on the dashboard
- histogram^{info} could be shown on a map, or a line chart, etc.

LAYOUT

time period variable, could be one year or between two years

Tornadoes in the US (2011)

Description about domain and time period choice

Map of all tornadoes



legends

text could go here too

text about the map

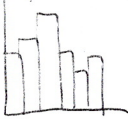
\$ damage vs deaths



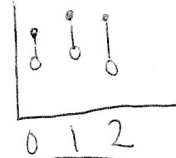
text could go here

text about scatter plot

Distribution of path length



Change over time for different EF ratings



text about the histogram and dot plot

FOCUS

interactive tooltip



Date: 01/01/2011
EF: 5
Deaths: 200
\$ damage: 1.7 million

Focus on the damage and severity of each tornado, and where they frequent most, hence the map.

OPERATIONS

- **time picker**: allows user to choose a time period to focus on, e.g. 2012-2013 (multiple years) or Feb 2011 (if looking at one year)
- **interactive tooltip**: hovering over each datapoint in every chart gives more info + exact numbers
- **Filterable legend**: click to filter:
(legend can vary)
 - EF rating (maybe)
 - Deaths (bins)
- **State picker**: allows focus on a state/states

DISCUSSION

- useful, informative map. However, not sure if it will get too complex with all the legends and filters applied.
- the rest of the charts are pretty simple, I probably could make them more detailed
- maybe add a time series, but then that could render the time picker useless
- layout is like a vertical dashboard, nice + simple
- could I look at other characteristics aside from those already in the viz?

Author: Amaris Zhang

Date: 17/9/2025

Sheet: 2

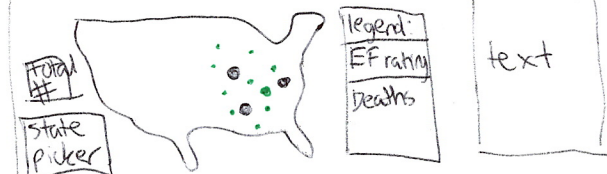
Task: Design #1, a vertical dashboard

LAYOUT

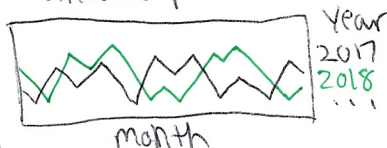
Tornadoes in the US (2017-2021)

Introductory paragraph

Map of all tornadoes



Tornadoes per month

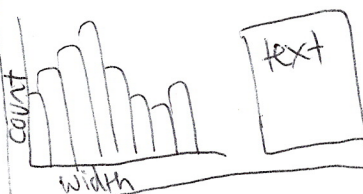


name + sources

\$ damage vs path length



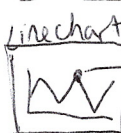
Distribution of width



FOCUS



Date:
EF:
Deaths:
Damage:
Location:



Year:
Tornadoes:
Total damage:
Total deaths:



Path length:
Damage:
Location:

Essentially same focus as sheet 2. Each chart will have an interactive tooltip to give more info.

OPERATIONS

- State picker: allows user to focus on one state/s
 - EF rating + deaths picker: focus on certain EF rating + death bins
 - Year picker (line chart): focus on one year / years
- These filters all change every chart's data
Every chart also has a "hover over" interactive tooltip

DISCUSSION

- Probably no scrolling needed
- Clear sections
- balance of charts + text
- not crowded, good use of white space
- good number of filters
- time series is useful

Author: Amaris Zhang

Date: 18/9/2025

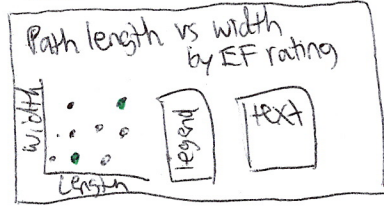
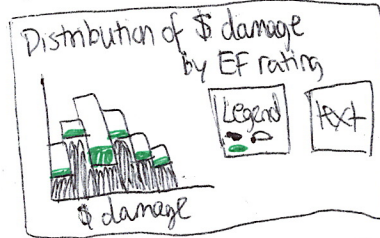
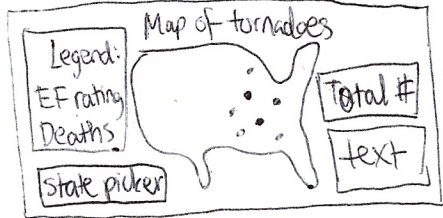
Sheet: 3

Task: Design #2, a horizontal, no-scroll dashboard

LAYOUT

Author + sources

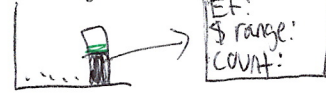
Fatal Tornadoes in USA (2017-2021)



FOCUS

Same focus as sheet 2, with more exploration in the damage + tornado L/W by EF rating.

Histogram



Scatterplot



OPERATIONS (map)

- state picker to focus on one state/multiple states
- year picker (area chart) to focus on a year/years
- interactive tooltip: hover over each datapoint in every chart for more info
- (maybe) death # picker: focus on tornadoes with a certain # of deaths, especially if I choose to include tornadoes w/o deaths

DISCUSSION

- each chart in its own clear section
- each chart is multi-dimensional, which gives more info
- area chart might be hard to compare between years visually
- consider stacked density plots instead of stacked histogram
- maybe change the legend of either the histogram/scatterplot to another attribute instead of EF (not many qualitative attributes to choose from though)
- could I add just one more chart? bar chart, pie chart, heatmap?
- not too much of a distinct design from sheet 3

Author: Amanis Zhang

Date: 19/9/2025

Sheet: 4

Task: Design #3, another more refined no-scroll dashboard

LAYOUT

Tornadoes in the USA (2020-2024)

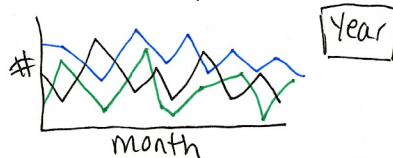
Intro text

Map of tornadoes



Text about Map

Tornadoes per month



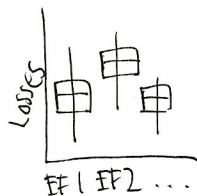
Text about line chart

Track length vs width



text about
scatterplot

Text about
boxplot



Author, sources, date

FOCUS

To find out the damage these tornadoes cause (by EF rating), and find any patterns. still same focus as before essentially, shown through interactive tooltip.

OPERATIONS

- state picker (map)
- Non-fatal/fatal tornadoes filter (map)
- Year selector (line chart)
- interactive tooltip via hovering over data points

DETAIL

1. combine 2 data sources + clean data, with excel
 2. create charts w/ vegalite + VS code
 3. add interactive elements w/ vegalite
 4. connect the charts via filters, with js
 5. create visualisation webpage w/ html, css, js and html. upload to Github
- Estimate everything will take a week to do

Author: Amaris Zhang
Date: 19/9/2025
Sheet: 5
Task: final visualisation design